ENTERPRISE AIOPS PLATFORM • ARCHITECTURE SPECIFICATION

Ingestion Best Practices, SLIs & Watermarking

Target Platform: Google Cloud Platform (GCP)
System of Record: ServiceNow ITSM
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AIOps Ingestion Pipeline: Best Practices

This guide outlines the best practices for designing, configuring, and operating the AIOps ingestion pipelines based on the core architecture defined in ingestion_architecture.md. Adhering to these guidelines ensures scalability, security, and high reliability across multi-cloud observability sources.

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1. Design Best Practices

1.1 Connector Pattern Selection

Always select the ingestion connector pattern based on the source's capabilities and network location:

- **Use Native Connectors (Cloud Logging/Log Router Sinks)** for any workloads already running on Google Cloud (e.g., GKE, Dataflow). This guarantees sub-second latency and zero egress costs.
- **Use Push Connectors (Cloud Run)** for modern SaaS platforms (Akamai, Dynatrace, Adobe, Splunk HEC) that support real-time webhooks or streaming exports.
- **Use Pull Connectors (Cloud Composer / Apache Airflow)** *only* for legacy on-premise databases or air-gapped systems that cannot push data out. Avoid using Cloud Run jobs for Pull scenarios, as Airflow provides superior retry logic and operational visibility for complex extraction tasks.

1.2 Strict Decoupling

- **Never connect sources directly to processing engines.** All incoming telemetry must land in **Cloud Pub/Sub** first.
- Use isolated, dedicated Pub/Sub topics for each major data source (e.g., `telemetry.akamai.raw`, `telemetry.splunk.raw`). This prevents a traffic flood from one source (like a DDoS attack logged by Akamai) from starving resources for other critical telemetry.

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2. Configuration Best Practices

2.1 Security & Authentication

- **Never hardcode credentials.** Cloud Run push connectors and Cloud Composer pull tasks must retrieve API keys, HMAC secrets, and database credentials dynamically from **GCP Secret Manager** at runtime.

- Protect all public-facing Cloud Run push connectors with **Cloud Armor WAF**. Enforce strict IP allowlisting (e.g., only accepting traffic from known Akamai or Dynatrace CIDR blocks) and implement rate-limiting to prevent malicious flooding.

2.2 Schema Management

- Enable and enforce the **Pub/Sub Schema Registry** using Protobuf or Avro.
- Connectors must validate JSON payloads against the registry before publishing. This prevents upstream SaaS providers from pushing undocumented, breaking schema changes that crash downstream Dataflow pipelines.

2.3 Resiliency Configuration

- **Multi-Region Pub/Sub**: Configure critical telemetry topics for cross-region message routing so that data ingestion survives single-zone or single-region outages.
- **Dataflow Auto-scaling**: Configure Dataflow streaming jobs with a generous `maxNumWorkers` limit. This allows the pipeline to gracefully absorb sudden spikes in error logs during a massive P1 outage.

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3. Operational & Observability Best Practices

3.1 Dead-Letter Queues (DLQs)

- **Never silently drop data.**
- Configure Pub/Sub native DLQs (after 5 failed delivery attempts) to catch messages that repeatedly crash Dataflow workers.
- Configure Dataflow side-outputs to catch messages that parse successfully but fail business logic validation (e.g., missing timestamps).
- Route all DLQ streams to a dedicated **Cloud Storage (GCS) Bucket** partitioned by date (`gs://aiops-dlq-bucket/YYYY/MM/DD/`) so data engineers can inspect and replay the data later.

3.2 Key Service Level Indicators (SLIs) to Monitor

You must configure **Cloud Monitoring** dashboards and alerts for the following critical ingestion SLIs:

- **Pub/Sub `oldest_unacked_message_age`**: If this exceeds 5 minutes, it indicates the Dataflow pipeline is stalled or falling behind.
- **Dataflow `system_lag` and `data_watermark_age`**: Monitors the real-time processing delay.
- **DLQ Bucket Depth**: A sudden spike in DLQ volume indicates an upstream schema change or a bad code deployment in the Dataflow parsing logic.

3.3 Incident Alerting Routing

- Route all critical ingestion pipeline alerts directly to **ServiceNow** via Cloud Monitoring webhooks.
- If the AIOps platform itself is failing to ingest data, the internal Data Platform SRE team must be paged immediately, as "silent monitoring failures" are the highest risk to enterprise uptime.